

The Force of Impact on a Cone Striking a Water Surface (Vertical Entry)

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Introduction

This paper contains work relating to the vertical impact of a cone on a water surface which was carried out at the Applied Mathematics Group, New York University, during the year 1945. If a cone enters water vertically at constant speed, similitude of the flow at different instants makes it possible to determine the force of impact in an exact manner by means of an iteration scheme involving an integral equation. This computation for a cone of vertex angle 120° was presented to Dr. A. Hillman, then with the Mathematical Tables Project, who perfected a method for carrying it out. The results of Dr. Hillman's work are summarized in a report of the Mathematical Tables Project dated January, 1946.

In the first section we develop an exact theory for the vertical entry of a cone at constant speed under the assumption that the fluid is an ideal incompressible one, and we present the results of Dr. Hillman's computation for the 120° cone. Since the computation is difficult, it seems unlikely that it will be repeated for various cone angles and, for this reason, it is desirable to give a simpler approximate theory from which the impact force can easily be computed for any cone angle. This is done in section 2 by taking, as first approximation, the velocity potential arising from an ellipsoid circumscribed about the cone and then making wetting and free surface corrections in the manner described in the report [2a] for the sphere.

Finally we compare our results with the experimental measurements of the Japanese investigator Watanabe [5a,5b].

Final results are presented in the form of a graph (Graph 1) which gives the dimensionless virtual mass k of the fluid as a function of the vertex angle θ of the cone. The actual virtual mass M is

$$M = k\sigma B^3 \tan^3 \frac{\theta}{2}$$

where σ is the fluid density and B is the depth of penetration below the initial water level. The impact force P is computed using the formula

$$P = \frac{3M}{B[1 + (M/M_0)]^3} U_0^2$$

where M_0 is the mass of the cone and U_0 the initial entry velocity.

*This paper was prepared while both authors were at New York University.

The curve of Graph 1 gives values of k determined from the approximate theory of section 2. The value of k computed by Hillman for the 120° cone as well as the values of k determined from the experiments of Watanabe are shown. Agreement is seen to be good.

1. Exact Theory

1.1. *Similitude.* Let the cone be at rest, and let the water surface rise vertically at constant speed. After making contact with the cone, the water surface is no longer plane but resembles Figure 1. Owing to the rotational symmetry it is necessary to draw only half of the section through the vertical axis. For particles on the free surface, introduce a Lagrangian coordinate s . This coordinate s is the distance which the given particle p had from the vertex of the cone at the

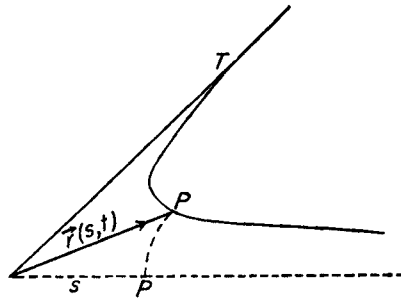


FIGURE 1

initial constant $t = 0$ which is the moment when the plane water surface touches the cone only at its vertex. Further, let $\mathbf{r} = \mathbf{r}(s, t)$ be the vector extending from the vertex of the cone to the particle s at the instant t . (Figure 1).

It is obvious that we have the law of similitude (compare [4])

$$(1.1.1) \quad \mathbf{r} = t \mathbf{R}(\xi)$$

where $\xi = s/t$. Moreover, on the free surface we have the boundary condition

$$(1.1.2) \quad \mathbf{r}_{tt} \cdot \mathbf{r}_s = 0.$$

Here the subscripts denote differentiation. Since $\mathbf{r}_s$ has the direction of the tangent to the surface, (1.1.2) states that the acceleration of a particle on the surface is normal to the surface. In other words, (1.1.2) is equivalent to the assertion that the pressure is constant over the free surface.

The relations (1.1.1) and (1.1.2) imply a third, namely that s , the coordinate of P , is equal to the distance measured along the surface at time t from the particle P to the point T where the free surface meets the cone. Thus the arc length along the surface from the cone to any given particle is independent of

the time. This condition, which we call the arc-length condition, may be expressed by the equation

$$(1.1.3) \quad \mathbf{r}_s^2 = 1.$$

To derive (1.1.3), we differentiate (1.1.1) with respect to t , and we find that

$$(1.1.4) \quad \mathbf{r}_t = \mathbf{R}(\xi) - \xi \mathbf{R}'(\xi),$$

$$(1.1.5) \quad \mathbf{r}_{tt} = \frac{\xi^2}{t} \mathbf{R}''(\xi).$$

Differentiating (1.1.1) with respect to s , we have also

$$(1.1.6) \quad \mathbf{r}_s = \mathbf{R}'(\xi).$$

Thus, multiplying both sides of (1.1.4) by t and using (1.1.6), we obtain

$$(1.1.7) \quad t\mathbf{r}_t = \mathbf{r} - s\mathbf{r}_s.$$

Substituting from (1.1.5) and (1.1.6) into the boundary condition (1.1.2), we have

$$(1.1.8) \quad \mathbf{R}''(\xi) \cdot \mathbf{R}'(\xi) = \frac{1}{2} \frac{d}{d\xi} \{\mathbf{R}'(\xi)\}^2 = 0.$$

Thus

$$\mathbf{r}_s^2 = \{\mathbf{R}'(\xi)\}^2 = c^2$$

where c is independent of ξ . But at $\xi = \infty$ it is clear that $\mathbf{r}_s^2 = 1$, so (1.1.3) is valid along the entire free surface.

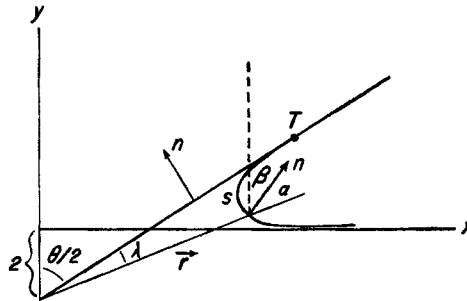


FIGURE 2

Now let the normal vector n point into the exterior of the fluid, and let the angle between the normal vector and the vector $\mathbf{r}$ be α . The angle between the normal and the vertical direction will be called β . Taking the scalar product of (1.1.7) with the unit normal and unit tangent directions, we find

$$(1.1.9) \quad w_n = r \cos \alpha$$

$$w_s = r \sin \alpha - s$$

where v_n and v_t are the velocity components normal and tangential to the free surface and where r is the length of the vector r .

Suppose that the water is at rest at infinity and that the cone is moving downward with constant speed U_0 . For simplicity we set $U_0 = 1$ and $t = 1$. The situation is then as shown in Figure 2 where the x -axis represents the original water level. Let φ be the velocity potential and ψ the stream function of the flow. We adopt the convention that the velocity at a point of the fluid is the negative gradient of φ . At points of the free surface we have at once from (1.1.9)

$$(1.1.10) \quad -\frac{\partial \varphi}{\partial n} = r \cos \alpha - \cos \beta = \frac{1}{x} \frac{\partial \psi}{\partial s}$$

$$-\frac{\partial \varphi}{\partial s} = r \sin \alpha - s - \sin \beta.$$

We observe that

$$(1.1.11) \quad \cos \alpha \, ds = r \, d\lambda, \quad \sin \alpha \, ds = dr, \quad \cos \beta \, ds = dx.$$

Moreover, since the angle λ between the vector $\mathbf{v}$ and the cone is given by

$$\lambda = \arctan \left(\frac{x}{1+y} \right) + \text{constant},$$

we have

$$(1.1.12) \quad d\lambda = \frac{1}{r^2} \{ (1+y) \, dx - x \, dy \}.$$

Substituting from (1.1.11) and (1.1.12) into (1.1.10), we find that

$$(1.1.13) \quad d\psi = x(r^2 \, d\lambda - dx) = x(y \, dx - x \, dy)$$

$$d\varphi = s \, ds - r \, dr + dy$$

at points of the free surface. Since $r^2 = x^2 + (1+y)^2$, integration gives

$$2\varphi = s^2 - r^2 + 2y + \text{constant}$$

$$= s^2 - x^2 - (1+y)^2 + 2y + \text{constant}$$

$$= s^2 - x^2 - y^2 + \text{constant}.$$

Since a constant may always be subtracted from a potential function, we have finally at points of the free surface

$$(1.1.14) \quad 2\varphi = s^2 - x^2 - y^2$$

$$d\psi = x(y \, dx - x \, dy).$$

At points of the cone

$$(1.1.15) \quad -\frac{\partial \varphi}{\partial n} = -\sin \frac{\theta}{2} = \frac{1}{x} \frac{\partial \psi}{\partial s}$$

where $\theta/2$ is the semi-angle at the vertex and s denotes arc-length along the cone. Hence

$$d\psi = -x \sin \frac{\theta}{2} ds = -x dx$$

and, since an additive constant may always be subtracted, we see that

$$(1.1.16) \quad \psi = -\frac{1}{2}x^2$$

at points of the cone which are wetted by the water.

Since $x = (y + 1) \tan \theta/2$ on the straight segment of the cone, we see that

$$x(y dx - x dy) = x \left[\left(\cot \frac{\theta}{2} x - 1 \right) dx - \cot \frac{\theta}{2} x dx \right] = -x dx$$

there. Therefore

$$(1.1.17) \quad d\psi = x(y dx - x dy)$$

on the whole of the boundary of the fluid.

For computational purposes it is better to express the arc-length condition (1.1.3) in the form

$$(1.1.18) \quad x_\tau = \int_0^{v(\tau)} \left(\frac{ds}{dy} + \frac{dx}{dy} \right) dy \quad (\text{arc-length condition}).$$

Here $x = x(y)$ is the curve which defines the shape of the free surface. Since the volume of water above the initial surface is equal to the volume of the cone below the initial surface, we have also

$$\pi \int_0^{v(\tau)} x^2 dy - \pi \int_0^{v(\tau)} \tan^2 \frac{\theta}{2} (1 + y)^2 dy = \pi \int_{-1}^0 \tan^2 \frac{\theta}{2} (1 + y)^2 dy;$$

that is

$$(1.1.19) \quad \int_0^{v(\tau)} x^2 dy = \frac{\tan^2 \theta/2}{3} (1 + y_\tau)^3 \quad (\text{volume condition}).$$

Thirdly, near infinity the leading term in the expansion of the velocity potential is a dipolar term. For large x the velocity of a particle on the free surface is therefore of the order of magnitude $1/x^3$, and it follows that the height of the free surface above the initial level is of the order of magnitude $1/x^3$ for large x . That is

$$(1.1.20) \quad \lim_{v \rightarrow 0} yx^3 > 0 \quad (\text{asymptotic condition}).$$

Finally, the surface curve is tangent to the cone at the point T (tangency condition). These four conditions relating to arc-length, volume, asymptotic behavior and tangency will be used in determining a first approximation to the curve which defines the free surface.

1.2. The ring source. If a shape for the free surface is assumed known, then the values of the velocity potential φ are determined everywhere on the boundary of the fluid except for the rectilinear portion which corresponds to the cone. We may express the value of φ at an interior point p of the fluid in terms of its boundary values by means of Green's theorem. Letting the point p tend to a point on the straight-line segment of the cone, we obtain in the usual way an integral equation for the unknown boundary values of φ on the cone. In expressing φ by means of its boundary values, we require a fundamental solution of the equation

$$(1.2.1) \quad \operatorname{div} (x \operatorname{grad} V) = x \frac{\partial^2 V}{\partial x^2} + x \frac{\partial^2 V}{\partial y^2} + \frac{\partial \varphi}{\partial x} = 0.$$

A point in three-dimensional space is expressed in terms of the coordinates x, y and an additional coordinate τ which is an angle of rotation about the y -axis from some fixed plane section. In terms of these coordinates the Laplace equation is

$$(1.2.2) \quad \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial V}{\partial x} \right) + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{x^2 \partial \tau^2} = 0,$$

and (1.2.1) is obtained from this equation by omitting the term which involves differentiation with respect to τ . In other words, (1.2.1) is the equation satisfied by all rotationally symmetric potential functions. Let $p(x, y, \tau)$ and $q(x_0, y_0, \tau_0)$ be two points in space. The distance between these two points is equal to

$$[x^2 + x_0^2 - 2xx_0 \cos(\tau - \tau_0) + (y - y_0)^2]^{1/2},$$

and therefore

$$(1.2.3) \quad V(p, q) = -\frac{1}{2} \int_{\lambda_0=0}^{2\pi} \frac{d\lambda_0}{[x^2 + x_0^2 - 2xx_0 \cos(\lambda - \lambda_0) + (y - y_0)^2]^{1/2}}$$

is a fundamental solution of (1.2.1) which is called a ring source. We observe that the ring source has a logarithmic singularity at the point (x_0, y_0) , and it is therefore the analogue of the logarithmic pole in ordinary plane potential theory.

The theorem of Gauss states that

$$(1.2.4) \quad \iint [\operatorname{div} (V \cdot x \operatorname{grad} \varphi) - \operatorname{div} (\varphi \cdot x \operatorname{grad} V)] dx dy = \int \left(V x \frac{\partial \varphi}{\partial n} - \varphi x \frac{\partial V}{\partial n} \right) dt.$$

Here n denotes the normal pointing into the exterior of the region in the x, y -plane and dt is an element of arc-length on its boundary. In applying (1.2.4),

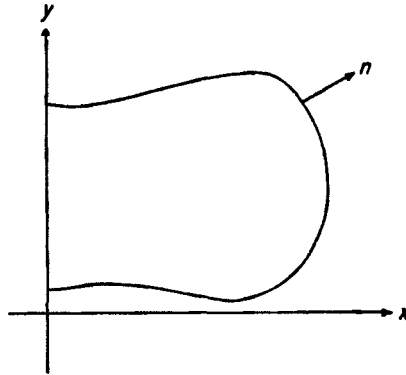


FIGURE 3

it is assumed that V and φ are without singularities in the region in question. Now let V be the ring source (1.2.3) and let φ be the velocity potential y ; assume that the point q lies in the region. We remove a small circle about q , and apply formula (1.2.4) to the remaining region. Since

$$(1.2.5) \quad \begin{aligned} \operatorname{div} (V \cdot x \operatorname{grad} \varphi) - \operatorname{div} (\varphi \cdot x \operatorname{grad} V) \\ = V \operatorname{div} (x \operatorname{grad} \varphi) - \varphi \operatorname{div} (x \operatorname{grad} V) = 0 \end{aligned}$$

there, we conclude in the usual way (on letting the radius of the circle about q shrink to zero) that

$$(1.2.6) \quad \varphi(p) = \frac{1}{2\pi} \int \left[\varphi(q)x(q) \frac{\partial V(p, q)}{\partial n_q} - V(p, q)x(q) \frac{\partial \varphi(p)}{\partial n_q} \right] dt_q.$$

To obtain a convenient form of $V(p, q)$ for computational purposes, we follow Trefftz [3] and express $V(p, q)$ in terms of an elliptic integral. Writing $\omega = \tau_0 - \tau$, we see from (1.2.3) that

$$(1.2.7) \quad \begin{aligned} V(p, q) &= -\frac{1}{2} \int_{\omega=0}^{2\pi} \frac{d\omega}{[x^2 + x_0^2 + (y - y_0)^2 - 2xx_0 \cos \omega]^{1/2}} \\ &= -\int_0^\pi \frac{d\omega}{[x^2 + x_0^2 + (y - y_0)^2 - 2xx_0 \cos \omega]^{1/2}}. \end{aligned}$$

Let

$$(1.2.8) \quad r_1^2 = (x - x_0)^2 + (y - y_0)^2, \quad r_2^2 = (x + x_0)^2 + (y - y_0)^2.$$

Then

$$(1.2.9) \quad x^2 + x_0^2 + (y - y_0)^2 = \frac{1}{2}(r_2^2 + r_1^2), \quad 2xx_0 = \frac{1}{2}(r_2^2 - r_1^2);$$

therefore

$$(1.2.10) \quad \begin{aligned} V(p, q) &= - \int_0^\pi \frac{d\omega}{[\frac{1}{2}(r_2^2 + r_1^2) - \frac{1}{2}(r_2^2 - r_1^2) \cos \omega]^{1/2}} \\ &= -2 \int_0^{\pi/2} \frac{d\sigma}{(r_1^2 \cos^2 \sigma + r_2^2 \sin^2 \sigma)^{1/2}} \end{aligned}$$

where $\sigma = \omega/2$. By the well-known Landen transformation

$$(1.2.11) \quad \rho = \rho(p, q) = \frac{1}{2}(r_2 + r_1), \quad \gamma = \gamma(p, q) = (r_1 r_2)^{1/2},$$

we then obtain

$$(1.2.12) \quad V(p, q) = -2 \int_0^{\pi/2} \frac{d\sigma}{(\rho^2 \cos^2 \sigma + \gamma^2 \sin^2 \sigma)^{1/2}}.$$

Writing

$$(1.2.13) \quad \xi = \frac{r_2 - r_1}{r_2 + r_1} = \frac{r_2^2 - r_1^2}{(r_2 + r_1)^2} = \frac{xx_0}{\rho^2},$$

we have finally

$$(1.2.14) \quad V(p, q) = - \frac{2K(\xi^2)}{\rho(p, q)}, \quad K(\xi^2) = \int_0^{\pi/2} \frac{d\sigma}{(1 - \xi^2 \sin^2 \sigma)^{1/2}}.$$

We also require an expression for the normal derivative of $V(p, \rho)$. We have

$$(1.2.15) \quad \frac{dK}{d\xi} = \frac{\xi}{1 - \xi^2} \left\{ \left(1 - \frac{1}{\xi^2}\right) K(\xi^2) + \frac{1}{\xi^2} E(\xi^2) \right\}$$

where

$$E(\xi^2) = \int_0^{\pi/2} (1 - \xi^2 \sin^2 \sigma)^{1/2} d\sigma.$$

Thus

$$(1.2.16) \quad \begin{aligned} \frac{\partial V}{\partial n_p} &= \left(\frac{\xi}{x_0}\right)^{1/2} \frac{1}{(x^3)^{1/2}} K(\xi^2) \frac{\partial x}{\partial n_p} \\ &- \left[\frac{1}{(xx_0\xi)^{1/2}} K(\xi^2) + 2\left(\frac{\xi}{xx_0}\right)^{1/2} \frac{\xi}{1 - \xi^2} \left\{ \left(1 - \frac{1}{\xi^2}\right) K(\xi^2) + \frac{1}{\xi^2} E(\xi^2) \right\} \right] \frac{\partial \xi}{\partial n_p}. \end{aligned}$$

Let α be the angle indicated in Figure 4. Then

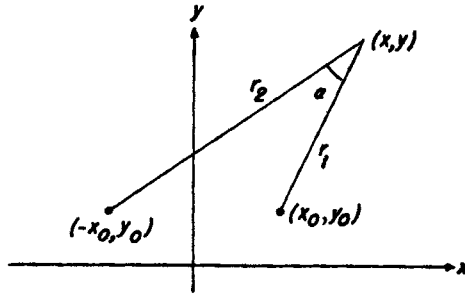


FIGURE 4

$$\frac{\partial \log (r_1/r_2)}{\partial n} = \frac{\partial \alpha}{\partial t}.$$

Since

$$\frac{r_1}{r_2} = \frac{1 - \xi}{1 + \xi},$$

we have

$$\frac{1 + \xi}{1 - \xi} \frac{\partial}{\partial n} \left(\frac{1 - \xi}{1 + \xi} \right) = -2 \frac{1}{1 - \xi^2} \frac{\partial \xi}{\partial n} = \frac{\partial \alpha}{\partial t},$$

that is

$$(1.2.17) \quad \frac{\partial \xi}{\partial n} = - \frac{1 - \xi^2}{2} \frac{\partial \alpha}{\partial t}.$$

Also

$$(1.2.18) \quad \frac{\partial x}{\partial n} = \frac{\partial y}{\partial t}.$$

Writing

$$(1.2.19) \quad G(\xi^2) = E(\xi^2) - \frac{1 - \xi^2}{2} K(\xi^2),$$

we obtain the formula

$$(1.2.20) \quad x \frac{\partial V}{\partial n_p} = \frac{1}{\rho} K(\xi^2) \frac{\partial y}{\partial t} + \frac{\rho}{x} G(\xi^2) \frac{\partial \alpha}{\partial t}.$$

By (1.2.14), this may also be written as

$$(1.2.20)' \quad x \frac{\partial V}{\partial n_p} = - \frac{1}{2} V \frac{\partial y}{\partial t} + \frac{\rho}{x} G \frac{\partial \alpha}{\partial t}.$$

We have

$$(1.2.21) \quad d\psi = x(q) \frac{\partial \varphi}{\partial n_a} dt_a$$

and, if $S = S(p, q)$ denotes the stream function corresponding to $V(p, q)$,

$$(1.2.22) \quad dS = x(q) \frac{\partial V(p, q)}{\partial n_a} dt_a.$$

Equation (1.2.6) may therefore be written in the form

$$(1.2.23) \quad 2\pi\varphi(p) = \int [\varphi(q) dS(p, q) - V(p, q) d\psi(q)]$$

where

$$(1.2.24) \quad dS(p, q) = \frac{1}{x_p} \rho G(\xi^2) d\alpha_a + \frac{K(\xi^2)}{\rho} dy_a = \frac{1}{x_p} \rho G(\xi^2) d\alpha_a - \frac{1}{2} V(p, q) dy_a.$$

Let, now, in equation (1.2.23), p tend to a boundary point which we also denote by p , and let I_p be a small arc of the boundary centered on p . Since V has only a logarithmic singularity, we see that the integrals

$$\int_{I_p} V d\psi, \quad \int_{I_p} \varphi V dy$$

tend to zero as the length of the interval shrinks to zero. However, as the variable point q approaches p ,

$$\rho(p, q) \rightarrow x_0, \quad \xi \rightarrow 1, \quad G(\xi^2) \rightarrow 1, \quad \varphi(q) \rightarrow \varphi(p);$$

therefore

$$\int_{I_p} \varphi(q) \frac{1}{x_q} \rho G(\xi^2) d\alpha_a \rightarrow \varphi(p) \int_{I_p} d\alpha_a = \beta\varphi(p)$$

where β is the exterior angle of the boundary at p . In particular, if the boundary curve has a tangent at p , then $\alpha = \pi$. Hence at regular points of the boundary we have the integral equation

$$(1.2.25) \quad \varphi(p) = \frac{1}{\pi} \int [\varphi(q) dS(p, q) - V(p, q) d\psi(q)].$$

Here the integration is to be taken over the boundary of the fluid in the sense in which the fluid lies to the left of the direction of integration (positive sense).

1.3. Virtual mass and impact force. Let U be the downward velocity of the cone at any instant t , and let P be the resultant upward impact force exerted on the cone. Then the cone exerts a downward force P on the fluid and imparts

momentum to the fluid. The virtual mass M of the fluid at a given instant t is defined by

$$(1.3.1) \quad P = \frac{d}{dt}(MU) \quad \text{or} \quad MU = \int_0^t P \, dt.$$

Thus MU is the vertical impulse contributed to the fluid by the moving cone.

Let M_0 be the mass of the incoming cone. Neglecting gravity compared to the impact force P , we have

$$(1.3.2) \quad -P = \frac{d}{dt}(M_0 U).$$

Adding (1.3.1) and (1.3.2) we obtain

$$\frac{d}{dt}[(M + M_0)U] = 0,$$

and integrating

$$(1.3.3) \quad (M + M_0)U = M_0 U_0 \quad \text{or} \quad U = \frac{U_0}{1 + (M/M_0)}$$

where U_0 is the velocity of the incoming cone at the initial instant when its vertex makes contact with the fluid.

Let $B = B(t)$ be the depth at time t of the vertex of the cone below the initial surface level. Then $c = c(t)$, the radius of the circle of intersection of the cone with the initial surface, is given by

$$(1.3.4) \quad c = B \tan \frac{\theta}{2}$$

where θ is the vertex angle of the cone. The velocity U of the cone is then $\dot{B} = dB/dt$; by differentiating (1.3.3) with respect to t and using (1.3.2), it follows that

$$(1.3.5) \quad P = \frac{dM/dB}{1 + (M/M_0)} U^2 = \frac{dM/dB}{[1 + (M/M_0)]^3} U_0^2.$$

We assume that the mass of the cone is large compared to the mass of water displaced, then M/M_0 may be neglected. Under these circumstances we have $U = U_0$ by (1.3.3)—that is, the cone enters the water at constant velocity—and (1.3.5) becomes

$$(1.3.5)' \quad P = \frac{dM}{dB} U_0^2.$$

Let

$$(1.3.6) \quad M = k\sigma c^3 = k\sigma B^3 \tan^3 \frac{\theta}{2}$$

where σ is the density of the fluid. Because of the similitude, the dimensionless factor k is a constant. Substituting from (1.3.6) into (1.3.5)', we obtain

$$(1.3.7) \quad P = 3k\sigma \tan^3 \frac{\theta}{2} B^2 \cdot U_0^2 = 3k\sigma \tan^3 \frac{\theta}{2} U_0^4 t^2$$

where t is the time measured from the instant when the vertex meets the water surface.

To compute the virtual mass M , we use the principle of conservation of momentum. The total momentum of the fluid is directed vertically downward because of the rotational symmetry and is equal to

$$(1.3.8) \quad 2\pi \iint \sigma \frac{\partial \varphi}{\partial y} x \, dx \, dy = 2\pi\sigma \int_{W+S} \varphi x \, dx - 2\pi\sigma \int_{H_\infty} \varphi x \, dx.$$

Here W is the wetted portion of the cone, S is the free boundary, and the last term in (1.3.8) means the limit of the integral extended over the submerged portion H of a large spherical surface the radius of which tends to infinity.

The external forces acting on the fluid contained in the membrane H are: (i) the downward force of magnitude P exerted by the sphere and (ii) the upward force exerted on H by the pressure of the fluid external to it. The resultant force due to the constant atmospheric pressure is zero, and the only effect of gravity is the buoyancy which is negligible compared to the impact force P . The force (ii) is equal to the time rate of change of the last term in (1.3.8). For, by Bernoulli's law, the force (ii) differs from the time rate of change of the last term by a term which is an integral of the square of the fluid velocity over H ; and this tends to zero as H approaches infinity. Likewise the momentum flux through H consists of integrals of squares of velocities, and these approach zero as H becomes infinite. Since the resultant of all the external forces is equal to the time rate of change of the momentum (1.3.8), we therefore have

$$(1.3.9) \quad P = \frac{d}{dt} \left(2\pi\sigma \int_{W+S} \varphi x \, dx \right).$$

Assume without loss of generality that $U_0 = 1$. Then

$$(1.3.10) \quad M = 2\pi\sigma \int_{W+S} \varphi x \, dx.$$

Owing to the form of the free boundary, it is convenient to express the integration over S in (1.3.10) in terms of y . Taking into account the proper signs, we have

$$(1.3.11) \quad \int_S \varphi x \, dx = - \int_0^{y^{(T)}} \varphi \frac{dx}{dy} dy.$$

Thus

$$(1.3.10)' \quad M = 2\pi\sigma \left\{ \int_0^{x^{(T)}} \varphi x \, dx - \int_0^{y^{(T)}} \varphi \frac{dx}{dy} dy \right\}$$

where the first integral refers to the boundary of the cone and the second to the free boundary. Finally, if $t = 1$ (in which case $B = 1$),

$$(1.3.12) \quad k = M \cot^3 \frac{\theta}{2}.$$

Instead of deriving the force of impact from momentum considerations, we may derive it from energy considerations (see [2b], Appendix). Let $\frac{1}{2}M_*U^2$ be the kinetic energy of the fluid at time t , where M_* is the "energy" virtual mass. By the principle of conservation of energy, we have

$$(1.3.13) \quad \frac{1}{2}(M_* + M_0)U^2 = \frac{1}{2}M_0U_0^2, \quad \text{or} \quad U^2 = \frac{U_0^2}{1 + (M_*/M_0)}.$$

Differentiating with respect to t and using (1.3.2), we obtain from (1.3.13)

$$(1.3.14) \quad P = \frac{1}{2} \frac{dM_*/dB}{1 + (M_*/M_0)} U^2 = \frac{1}{2} \frac{dM_*/dB}{[1 + (M_*/M)]^2} U_0^2.$$

Comparison of formulas (1.3.3) and (1.3.13) shows that

$$(1.3.15) \quad 1 + \frac{M_*}{M_0} = \left(1 + \frac{M}{M_0}\right)^2 \quad \text{or} \quad M_* = 2M \left(1 + \frac{1}{2} \frac{M}{M_0}\right).$$

Hence for incoming cones of large mass compared to the mass of water displaced, we have

$$(1.3.16) \quad M_* = 2M.$$

In order to obtain an expression for M_* , we observe that the total kinetic energy of the fluid is

$$\pi\sigma \iint (\text{grad } \varphi)^2 x \, dx \, dy = -\pi\sigma \int \varphi \frac{\partial \varphi}{\partial n} x \, ds$$

where ds denotes an element of arc-length on the free boundary curve in the x, y -plane. Assuming that $U_0 = 1$, we therefore have

$$M = \frac{1}{2} M_* = -\pi\sigma \int \varphi \frac{\partial \varphi}{\partial n} x \, ds = \pi\sigma \int \varphi \frac{\partial \psi}{\partial s} ds = \pi\sigma \int \varphi \, d\psi$$

since

$$-\frac{\partial \varphi}{\partial n} = \frac{1}{x} \frac{\partial \psi}{\partial x}.$$

That is,

$$(1.3.17) \quad M = \frac{1}{2} M_* = \pi\sigma \int_{w+s} \varphi \, d\psi.$$

The problem of determining the impact force is thus solved if we know the free boundary and the values of φ on the straight line of the cone. For then φ and $d\psi$ are determined everywhere on the boundary of the fluid by (1.1.14) and (1.1.17), and we may compute M using either formula (1.3.10) or (1.3.17).

We can roughly describe the procedure as follows. Assume a form for the free boundary curve. Then $d\psi$ is determined everywhere on the boundary, and φ is determined on the free boundary curve. Equation (1.2.25) thus becomes an integral equation for the values of φ on the segment of the cone. Having determined the values of φ on the straight line from the integral equation, we substitute these values under the integral in (1.2.25) and take p on the free boundary. If the free boundary has been chosen correctly, the values of $\varphi(p)$ given by (1.2.25) will agree with the values $\varphi = \frac{1}{2}(s^2 - x^2 - y^2)$ given by (1.1.14). If the curve is not correct, we choose a second approximation to the curve and repeat the process. Dr. Hillman perfected a method for carrying out this difficult iteration process, and we now discuss the procedure which he applied to the cone with vertex angle 120° .

1.4. Calculation of the free boundary and virtual mass. The boundary of the fluid consists of a straight part along the line

$$(1.4.1) \quad x = \tan 60^\circ \cdot (y + 1) = 3^{1/2}(y + 1)$$

from $(0, -1)$ to the unknown point T to be designated by $(x(T), y(T))$, and an unknown curved part tangent to the straight line at T . On the curve let

$$(1.4.2) \quad x = x(y), \quad 0 \leq y \leq y(T).$$

On the whole boundary we have by (1.1.17)

$$(1.4.3) \quad d\psi = x(y) dx - x dy,$$

while on the curved part by (1.1.14)

$$(1.4.4) \quad \varphi = (s^2 - x^2 - y^2)/2$$

where s is the arc-length measured from T .

A method of solving the problem would be to approximate $x(y)$ by a curve depending on a finite number of parameters, and also to approximate φ by a function depending on a finite number of parameters. Then enough equations to solve for the parameters would be obtained by taking the point p in the integral equation (1.2.25) at different positions on the boundary. The difficulty with this method is the non-linear way in which $x(y)$ enters into the integral equation. This difficulty can be overcome if a sufficiently good approximation to $x(y)$ is known, for in this case $x(y)$ can be expressed in terms of parameters whose squares can be neglected.

On the straight part of the boundary the difference $y - y_p$ may be replaced by $(x - x_p)/3^{1/2}$, and the various functions may be expressed in terms of the

x -coordinates. If the point p lies on the straight part of the boundary, the equation (1.2.25) may be written in the form

$$(1.4.5) \quad \varphi(x_p) = F(x_p) - \frac{1}{\pi} \int_0^{x(T)} \varphi(x) S'(x_p, x) dx.$$

Here

$$(1.4.6) \quad S'(x_p, x) = \frac{\partial S(x_p, x)}{\partial x}$$

and

$$(1.4.7) \quad F(x_p) = \frac{1}{\pi} \int_{\text{curve}} (\varphi dS - V d\psi) - \frac{1}{\pi} \int_{\text{straight part}} V d\psi.$$

The kernel $S'(x_p, x)$ satisfies

$$(1.4.8) \quad S'(x_p, x) = S'(1, x/x_p)/x_p$$

where

$$(1.4.9) \quad S'(1, x) = - \frac{(x^2 + x + 1)^{1/2} + |x - 1|}{2(x^2 + x + 1)} G[\xi^2(x)] \\ + \frac{K[\xi^2(x)]}{(x^2 + x + 1)^{1/2} + |x - 1|},$$

$$(1.4.10) \quad \xi(x) = \frac{(x^2 + x + 1)^{1/2} - |x - 1|}{(x^2 + x + 1)^{1/2} + |x - 1|}.$$

We observe in particular that

$$(1.4.11) \quad S'(1, x) = S'(1, 1/x)/x.$$

Thus the kernel is completely expressible in terms of $S'(1, x)$ over the range $0 \leq x \leq 1$. $S'(1, x)$ has a logarithmic singularity at $x = 1$.

If the curve is assumed known, equation (1.4.5) becomes an integral equation. The integral can be replaced by a finite system of linear equations if φ is approximated by an expression of the form

$$(1.4.12) \quad \varphi = h_0(x) + c_1 h_1(x) + \cdots + c_n h_n(x)$$

where the $h_i(x)$ are piecewise polynomials and the c_i are unknown constants. It may be verified that

$$(1.4.13) \quad \varphi(0) = 2F(0)/3$$

where

$$(1.4.14) \quad F(0) = \frac{3^{1/2} x_t}{2} + \int_{\text{curve}} \left[\frac{d\psi}{\rho} - \frac{\varphi(d\psi + x dx)}{\rho^3} \right].$$

The method recommended by Hillman is one of linearization and iteration. First assume an approximate curve C_1 satisfying the following five conditions: (1) tangency condition—the curve is tangent to the straight line $x = 3^{1/2}(y + 1)$ at (x_T, y_T) ; (2) length condition—equation (1.1.18); (3) volume condition—equation (1.1.19); (4) asymptotic condition—equation (1.1.20); (5) near the point of tangency the curve is very close to the straight line $x = 3^{1/2}(y + 1)$. Conditions (4) and (5) suggest an equation of the form

$$(1.4.15) \quad x = cy^{-1/3} + \cdots + ky.$$

To make the volume condition linear in the coefficients, this can be replaced by

$$(1.4.16) \quad x^2 = a_{-2}y^{-2/3} + a_0 + a_2y^{2/3} + a_4y^{4/3} + a_6y^2.$$

Five terms are written, but more can be inserted if necessary. Choose the coordinates of three points on the curve, one of which should be the point of tangency $(x(T), y(T))$. The three chosen points and the volume and tangency conditions suffice to determine the coefficients. Since the curve is very straight near the point of tangency, the change in

$$\int_0^{y(T)} \left(\frac{ds}{dy} + \frac{dx}{dy} \right) dy$$

due to a variation of $(x(T), y(T))$ can be predicted to a fair degree of accuracy, thus enabling one to satisfy the length condition by proper choice of this point.

Having selected an approximate curve C_1 satisfying the above requirements, we can choose as parameters for the equation of the desired curve the differences in x -coordinates between the two curves C and C_1 for three definite values of y , one of which is $y(T)$. It may be more convenient to use differences in values of x^2 as parameters. These parameters (whichever have been chosen) will be assumed sufficiently small that their squares can be neglected (linearization). The length condition may be used to eliminate the parameter depending on $y(T)$; and when this is done the point of tangency will depend on the remaining parameters, say e_1 and e_2 . The function φ on the straight line may then be expressed in terms of e_1 and e_2 ; that is, the coefficients c_k in (1.4.12) may be expressed as linear functions of e_1 and e_2 . The final step in the first iteration is to solve for e_1 and e_2 using (1.2.25) for the points on the curve corresponding to $y = y_1$ and $y = y_2$. The e_1 and e_2 calculated in this way lead to a second approximation C_2 for the curve. The process may be repeated to obtain a third approximation, and so on.

It was suggested that a curve C_1 can be obtained by deforming

$$(1.4.17) \quad y = \frac{4}{\pi\lambda} [1 - (1 - \lambda^2)]^{1/2} - \frac{2}{\pi} \arcsin \lambda, \quad x = \frac{4(3)^{1/2}}{\pi\lambda}, \quad 0 \leq \lambda \leq 1,$$

so as to have the curve tangent to the line $y = x/3^{1/2} - 1$ at (4.022, 1.322). Dr. Hillman selected as a trial the curve

$$(1.4.18) \quad \begin{cases} y = \frac{4}{\pi\lambda} [1 - (1 - \lambda^2)^{1/2} - \frac{2}{\pi} \arcsin \lambda, x = \frac{4(3)^{1/2}}{\pi\lambda}, 0 \leq \lambda \leq 0.714 \\ x = 1.75531y + 1.67088 + 0.040899y^{-1}, 0.03 \leq y \leq 1.322. \end{cases}$$

A curve of the form (1.4.16) which is essentially one expression for the two pieces of (1.4.18) is

$$(1.4.19) \quad \begin{aligned} x^2 = & 0.6866584y^{-2/3} + 0.21950727 + 7.64383523y^{2/3} \\ & - 0.31368464y^{4/3} + 3.79548915y^2. \end{aligned}$$

The curve (1.4.19) is tangent to the line $x = 3^{1/2}(y + 1)$ at (4.02123195; 1.32165935).

For the curve (1.4.18) the values of φ on the straight part of the boundary were computed. Let

$$(1.4.20) \quad A(x_p) = \frac{1}{\pi} \int [\varphi(q) dS(p, q) - V(p, q) d\psi(q)]$$

where the integral is along the curve (1.4.18) from infinity to the point of tangency $(x(T), y(T))$ and where (x_p, y_p) is a point on the straight line $x = 3^{1/2}(y + 1)$, and let

$$(1.4.21) \quad B(x_p) = \frac{x_p}{\pi} \int_0^{4.02212/x_p} \frac{2(3)^{1/2} x K[\xi^2(x)] dx}{(x^2 + K + 1)^{1/2} + |x - 1|}$$

where $\xi(x)$ is given by (1.4.10). Then

$$(1.4.22) \quad F(x_p) = A(x_p) + B(x_p).$$

Graphs 2-5 are taken from Hillman's report [1]. The curve (1.4.18) is shown in Graph 2, and the various quantities entering into the solution of the integral equation (1.4.5) are plotted in Graph 3. Graphs 4 and 5 give additional information concerning values which enter, in particular, into the computation of the virtual mass.

Let

$$(1.4.23) \quad M_1 = \pi \int_{\text{boundary}} \varphi \frac{d\psi}{dy} dy$$

(formula (1.3.17) with $\sigma = 1$);

$$(1.4.24) \quad M_2 = 2\pi \left\{ \int_0^{x(T)} \varphi x dx - \int_0^{y(T)} \varphi \frac{dx}{dy} dy \right\}$$

(formula (1.3.10)' with $\sigma = 1$); and

$$(1.4.25) \quad M_3 = 2\pi \int_0^{y(T)} \varphi \left(\frac{d\psi}{dy} + x \frac{dx}{dy} \right) dy = 2M_1 - M_2.$$

For the curve (1.4.18) it was found that

$$\begin{aligned} M_1 &= 7.9 \\ (1.4.26) \quad M_2 &= 8.3 \\ M_3 &= 7.5 \end{aligned}$$

where there are indications that M_2 is closest to the true value. The value 8.3 for M gives

$$(1.4.27) \quad k = \frac{M}{B\sigma \tan^3(\theta/2)} = \frac{8.3}{(3^{1/2})^3} = 1.6$$

for the dimensionless virtual mass, and this value is indicated in Graph 1 (point indicated by a triangle).

2. Approximate Theory

2.1. *Approximation by the flow about an ellipsoid.* We circumscribe an ellipsoid about the cone as shown in Figure 5, and replace the entering cone at each

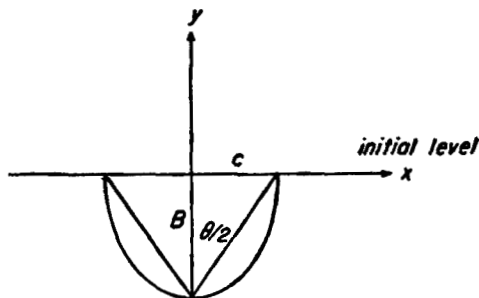


FIGURE 5

instant of time t by that portion of the ellipsoid which lies below the initial water level. As a first approximation we take the virtual mass M to be that due to the flow about this ellipsoid. The flow about an ellipsoid is classical. Let

$$(2.1.1) \quad \gamma = B/c, \quad \xi = c/x,$$

$$(2.1.2) \quad u^2 + 1 = \frac{1}{1 - \gamma^2} \cdot \frac{1}{\xi^2}, \quad 0 \leq \gamma < 1,$$

$$u^2 - 1 = \frac{1}{\gamma^2 - 1} \cdot \frac{1}{\xi^2}, \quad \gamma > 1,$$

and write

$$(2.1.3) \quad \mu = \begin{cases} \frac{2\gamma}{(1-\gamma^2)^{3/2}} \left[\frac{(1-\gamma^2)^{1/2}}{\gamma} - \arccos \gamma \right], & 0 < \gamma < 1 \\ \frac{2}{3}, & \gamma = 1 \\ \frac{2\gamma}{(\gamma^2-1)^{3/2}} \left[\operatorname{arccosh} \gamma - \frac{(\gamma^2-1)^{1/2}}{\gamma} \right], & \gamma > 1. \end{cases}$$

If $U = \dot{B} = dB/dt$ is the velocity, the potential φ of the ellipsoid flow is given by

$$(2.1.4) \quad \varphi = \begin{cases} -\frac{2Uy}{\pi} \left\{ \frac{\xi}{(1-\xi^2)^{1/2}} - \arcsin \xi \right\}, & \gamma = 0 \text{ (disc)} \\ -\frac{Uy}{2-\mu} \frac{2\gamma}{1-\gamma^2} \left\{ \frac{\xi}{[1-(1-\gamma^2)\xi^2]^{1/2}} - \frac{1}{(1-\gamma^2)^{1/2}} \arcsin (1-\gamma^2)^{1/2} \xi \right\}, & 0 < \gamma < 1 \\ -\frac{1}{2} Uy \xi^3, & \gamma = 1 \\ -\frac{Uy}{2-\mu} \frac{2\gamma}{\gamma^2-1} \left\{ \frac{1}{2} \frac{1}{(\gamma^2-1)^{1/2}} \cdot \log \frac{(1+(\gamma^2-1)\xi^2)^{1/2} + (\gamma^2-1)^{1/2}\xi}{(1+(\gamma^2-1)\xi^2)^{1/2} - (\gamma^2-1)^{1/2}\xi} - \frac{\xi}{[1+(\gamma^2-1)\xi^2]^{1/2}} \right\}, & \gamma > 1 \end{cases}$$

or

$$(2.1.4)' \quad \varphi = \begin{cases} -\frac{2Uy}{\pi} \left\{ \frac{1}{u} - \arctan \frac{1}{u} \right\}, & \gamma = 0 \\ -\frac{Uy}{2-\mu} \frac{2\gamma}{(1-\gamma^2)^{3/2}} \left\{ \frac{1}{u} - \arctan \frac{1}{u} \right\}, & 0 < \gamma < 1 \\ -\frac{1}{2} Uy \xi^3, & \gamma = 1 \\ -\frac{Uy}{2-\mu} \frac{2\gamma}{(\gamma^2-1)^{3/2}} \left\{ \operatorname{arctanh} \frac{1}{u} - \frac{1}{u} \right\}, & \gamma > 1. \end{cases}$$

We observe that the formulas for $\gamma > 1$ are derived from those for $\gamma < 1$ by analytic continuation around the point $\gamma = 1$ in the complex γ -plane.

Let M_E be the virtual mass of the ellipsoid flow (half the virtual mass of the full ellipsoid completely surrounded by fluid), and write

$$(2.1.5) \quad M_E = k_E \sigma c^3 = k_E \sigma B^3 \tan^3 \frac{\theta}{2}.$$

Then we find that

$$(2.1.6) \quad k_E = \frac{2\pi}{3} \gamma \frac{\mu}{2 - \mu}.$$

In Graph 6 the quantity k_E is exhibited as a function of the vertex angle θ of the cone. We observe the special cases

$$(2.1.7) \quad k_E = \begin{cases} \frac{4}{3}, & \gamma = 0, & \theta = \pi \\ \frac{\pi}{3}, & \gamma = 1, & \theta = \frac{\pi}{2} \\ 0, & \gamma = \infty, & \theta = 0. \end{cases}$$

2.2. The wetting correction. We now correct for the fact that the fluid rises in a neighborhood of the cone thus producing a greater effective depth of penetration into the water. Let

$$(2.2.1) \quad B_1 = wB$$

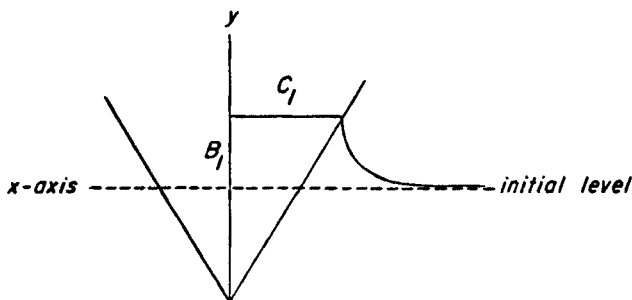


FIGURE 6

be the effective depth of penetration where w is a wetting factor (compare [2b]), and write

$$(2.2.2) \quad c_1 = wB \tan \frac{\theta}{2}.$$

To compute w , let $\eta(x)$ be the height above the initial level of the free surface at a point whose distance is x from the axis of symmetry. Assume that the upward velocity of the surface at the instant t is that induced by an ellipsoid of radius c_1 . Then

$$(2.2.3) \quad \dot{\eta} = - \left. \frac{\partial \varphi}{\partial y} \right|_{y=0} = \dot{B}F(\xi_1)$$

where $\xi_1 = c_1/x$. Because of the similitude, the factor w is independent of the time and we therefore obtain from (2.2.2) by differentiation

$$(2.2.4) \quad \dot{B} = \frac{\dot{c}_1}{w} \cot \frac{\theta}{2}.$$

Substituting from (2.2.4) into (2.2.3), we have

$$(2.2.5) \quad \dot{\eta} = \frac{\cot(\theta/2)}{w} F(\xi_1) \dot{c}_1.$$

Integrating (2.2.5) with respect to time, we find that the rise of the surface at the point is equal to

$$(2.2.6) \quad \eta = \eta(x) = \frac{\cot(\theta/2)}{w} \int_0^{c_1} F\left(\frac{c}{x}\right) dc = \frac{x \cot(\theta/2)}{w} \int_0^{\xi_1} F(\xi) d\xi.$$

By (2.2.2), the surface meets the cone when $c_1 = x$, that is when $\xi_1 = c_1/x = 1$. Thus

$$(2.2.7) \quad B_1 - B = B(w - 1) = \eta(c_1) = \frac{c_1 \cot(\theta/2)}{w} \int_0^1 F(\xi) d\xi = B \int_0^1 F(\xi) d\xi,$$

and we have

$$(2.2.8) \quad w - 1 = \int_0^1 F(\xi) d\xi.$$

We find that

$$(2.2.9) \quad w - 1 = \begin{cases} \frac{4}{\pi} - 1, & \gamma = 0 \\ \frac{2\gamma}{(2-\mu)(1-\gamma^2)^{3/2}} \cdot \left[\frac{2(1-\gamma)}{(1-\gamma^2)^{1/2}} - \arccos \gamma \right], & 0 < \gamma < 1 \\ \frac{1}{\gamma}, & \gamma = 1 \\ \frac{2\gamma}{(2-\mu)(\gamma^2-1)^{3/2}} \cdot \left[\operatorname{arccosh} \gamma - \frac{2(\gamma-1)}{(\gamma^2-1)^{1/2}} \right], & \gamma > 1 \\ 0, & \gamma = \infty. \end{cases}$$

By (1.3.10)

$$(2.2.10) \quad M = 2\pi\sigma \int_{w+s} \varphi x \, dx.$$

Writing

$$(2.2.11) \quad M_w = 2\pi\sigma \int_w \varphi x \, dx,$$

we have

$$(2.2.12) \quad M_w = k_w \sigma c^3 = k_w \sigma B^3 \tan^3 \frac{\theta}{2}$$

where

$$(2.2.13) \quad k_w = k_E \cdot w^3.$$

The quantity w is plotted as a function of vertex angle in Graph 7, and k_w is plotted in Graph 8.

2.3. *The free surface correction.* The free surface correction arises from the term

$$(2.3.1) \quad 2\pi\sigma \int_s \varphi x \, dx$$

occurring in (2.2.10) and the fact that we shall no longer neglect squares of velocity on S . The correction to the impact force P arises from the time derivative of (2.3.1) in which we assume that φ is the ellipsoid potential. Let the height of the surface above the initial level be $\eta = \eta(x, t)$, and denote the surface correction to the impact force by P_s . Then

$$(2.3.2) \quad P_s = 2\pi\sigma \frac{d}{dt} \int_s \varphi x \, dx = 2\pi\sigma \int_s \frac{d\varphi}{dt} x \, dx = 2\pi\sigma \int_s \left\{ \frac{\partial \varphi}{\partial t} + \dot{\eta} \frac{\partial \varphi}{\partial y} \right\} x \, dx$$

since $\varphi = 0$ on the surface in the case of the ellipsoid potential. The exact boundary condition on the free surface is that the pressure be constant, and therefore zero without loss of generality, since the resultant force due to a constant atmospheric pressure is zero. By Bernoulli's equation the exact boundary condition becomes

$$(2.3.3) \quad \frac{\partial \varphi}{\partial t} = \frac{1}{2} (\text{grad } \varphi)^2.$$

Substituting from (2.3.3) into (2.3.2) we obtain

$$(2.3.4) \quad P_s = 2\pi\sigma \int_s \left\{ \frac{1}{2} (\text{grad } \varphi)^2 + \dot{\eta} \frac{\partial \varphi}{\partial y} \right\} x \, dx = -\pi\sigma \int_s \left(\frac{\partial \varphi}{\partial y} \right)^2 x \, dx$$

since

$$\dot{\eta} = -\frac{\partial \varphi}{\partial y}, \quad (\text{grad } \varphi)^2 = \left(\frac{\partial \varphi}{\partial y}\right)^2.$$

Write

$$(2.3.5) \quad P_s = -3k_s \sigma \tan^3 \frac{\theta}{2} (B\dot{B})^2$$

where

$$(2.3.6) \quad \begin{aligned} k_s &= \frac{\pi \gamma^3}{3B^2} \int_s F(\xi) x \, dx = \frac{\pi \gamma^3 c_1^2}{3B^2} \int_1^0 F(\xi) \frac{1}{\xi} d\left(\frac{1}{\xi}\right) \\ &= \frac{\pi}{3} w^2 \gamma \int_0^1 F(\xi) \frac{d\xi}{\xi^3}. \end{aligned}$$

A computation shows that

$$(2.3.7) \quad k_s = \begin{cases} 0, & \gamma = 0 \\ \frac{\pi}{3} w^2 \gamma \left\{ \frac{\log \frac{1}{\gamma} - \frac{1-\gamma^2}{2}}{\left[\frac{\arccos \gamma}{(1-\gamma^2)^{1/2}} - \gamma \right]^2} - \frac{1}{2} \right\}, & 0 < \gamma < 1 \\ \frac{\pi}{3} \left(\frac{9}{8}\right)^2 \frac{1}{16}, & \gamma = 1 \\ \frac{\pi}{3} w^2 \gamma \left\{ \frac{\log \frac{1}{\gamma} - \frac{1-\gamma^2}{2}}{\left[\frac{\text{arccosh } \gamma}{(\gamma^2 - 1)^{1/2}} - \gamma \right]^2} - \frac{1}{2} \right\}, & \gamma > 1 \\ 0, & \gamma = \infty \end{cases}$$

Graph 9 shows k_s as a function of vertex angle θ .

Finally, by (1.3.1), (2.2.10), (2.2.11), (2.3.2) and (2.3.5), we have

$$P = M_w \ddot{B} + \dot{B}^2 \left(\frac{dM_w}{dB} - 3k_s \sigma \tan^3 \frac{\theta}{2} B^2 \right).$$

But

$$P = -M_0 \ddot{B}, \quad \dot{B} = \frac{U_0}{[1 + (M/M_0)]},$$

therefore

$$(2.3.8) \quad P = \frac{dM_w/dB - 3k_s \sigma \tan^3(\theta/2) B^2}{[1 + (M_w/M_0)][1 + (M/M_0)]^2} U_0^2$$

$$= \frac{dM/dB}{[1 + (M_w/M_0)][1 + (M/M_0)]^2} U_0^2$$

where

$$(2.3.9) \quad M = k\sigma c^3 = k\sigma \tan^3 \frac{\theta}{2} B^3, \quad k = k_w - k_s.$$

The quantity k is displayed in Graph 1 as a function of vertex angle θ .

In formula (2.3.8) the denominator is approximated by $(1 + M/M_0)^3$, so the final formula for P takes the form

$$(2.3.10) \quad P = \frac{\frac{dM}{dB}}{\left(1 + \frac{M}{M_0}\right)^3} U_0^2 = \frac{3k\sigma \tan^3 \frac{\theta}{2} B^2}{\left(1 + \frac{k\sigma \tan^3(\theta/2) B^3}{M_0}\right)} U_0^2.$$

The similitude of the flow at different times rests on the assumption that the entry velocity is constant. Therefore, strictly speaking, formula (2.3.10) should be used only when M_0 is large compared to the amount of water displaced, in which case (2.3.10) reduces to (1.3.7). However, formula (2.3.10) is a good approximation up to the depth of penetration where P attains its maximum P_m . The maximum P_m is seen to occur for

$$(2.3.11) \quad B = B_m = \cot \frac{\theta}{2} \left(\frac{2M_0}{7k\sigma} \right)^{1/3}.$$

3. Comparison with Experiment

3.1. *The measurements of Watanabe.* Careful experiments on the impact of cones striking a water surface vertically have been made by S. Watanabe (see [5a, 6]). He used cones on top of which were placed weights and piezoelectric gauges connected to an oscillograph. The system was dropped from various heights, and the oscillograph record gave the impact force as a function of the time.

From (2.3.10) we have approximately

$$(3.1.1) \quad P = \frac{3k\sigma \tan^3(\theta/2) B^2}{1 + \frac{3k\sigma \tan^3(\theta/2) B^3}{M_0}} U_0^2.$$

Solving (3.1.1) for k we obtain

$$(3.1.2) \quad k = \frac{P}{3\sigma \tan^3 \frac{\theta}{2} B^2 U_0^2 \left(1 - \frac{BP}{M_0 U_0^2}\right)}.$$

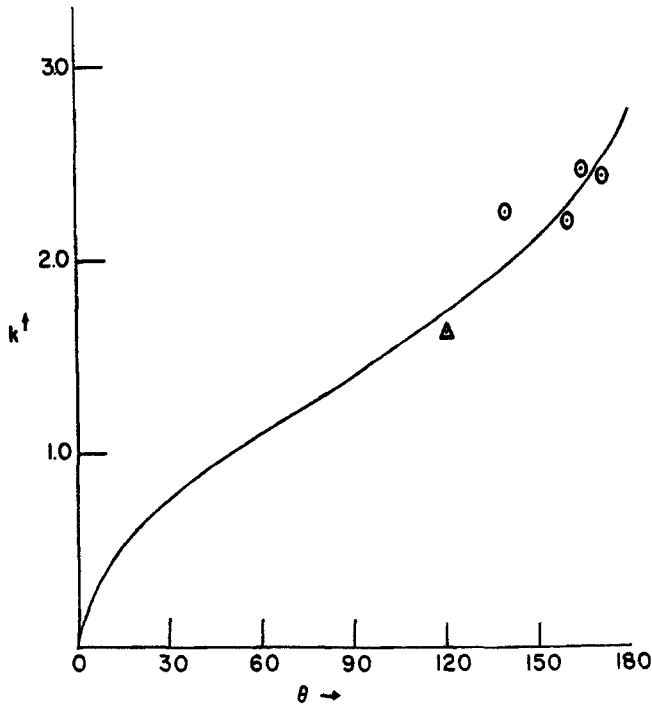
Let H be the height in centimeters from which the cone is dropped, R the impact force in kilograms at time t , B the depth of penetration of the cone in centimeters at time t , and M_0 the mass of the cone in kilograms. Expressed in terms of these quantities, formula (3.1.2) becomes

$$(3.1.3) \quad k = 10^3 \frac{R}{B^2} \cdot \frac{\cot^3 \frac{\theta}{2}}{6H} \cdot \frac{1}{1 - \frac{BR}{2M_0H}}.$$

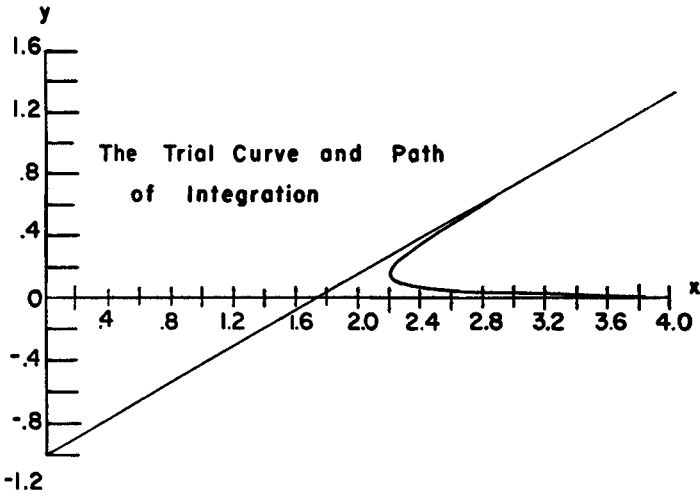
In the Appendix, measurements of these quantities obtained from [5b] are tabulated together with the corresponding values of k calculated from (3.1.3). Watanabe used cones with vertex angles 140° , 160° , 165° and 170° only; the average values of k determined from these measurements are:

Cone angle	140°	160°	165°	170°
k	2.24	2.21	2.45	2.43

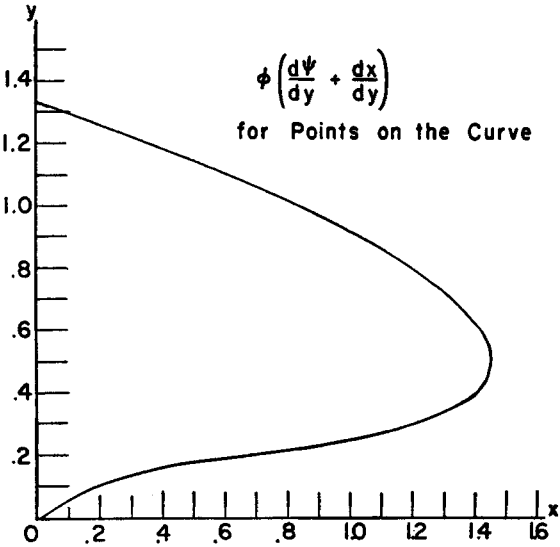
These values are circled in Graph 1.



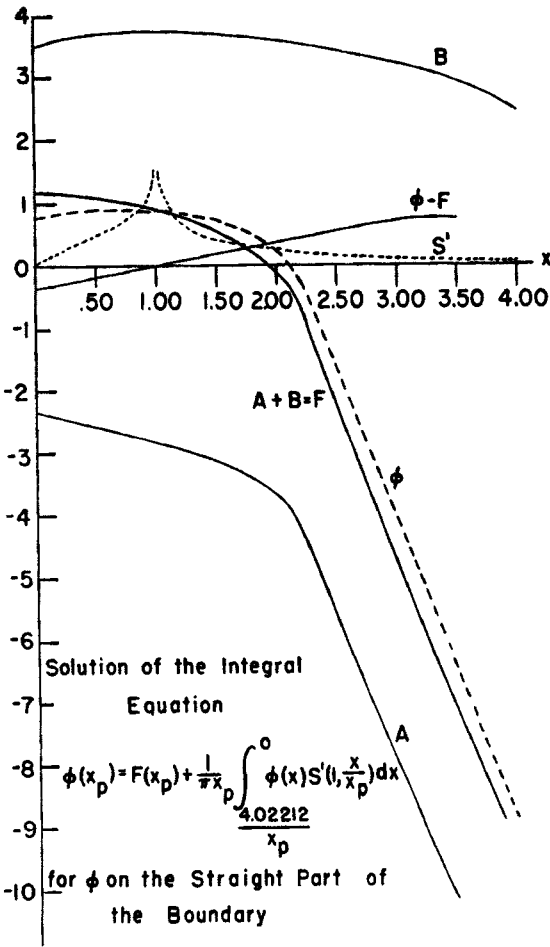
GRAPH 1: (○) Watanabe's experiments; (Δ) Hillman's computations.



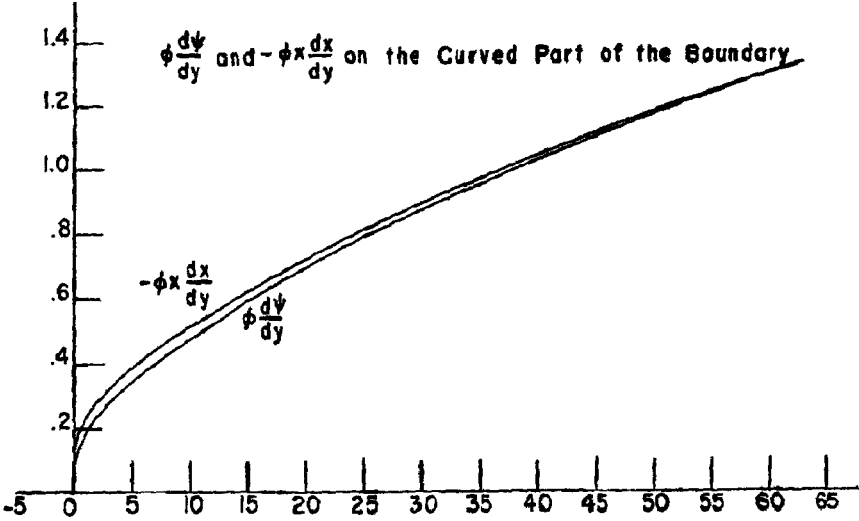
GRAPH 2a



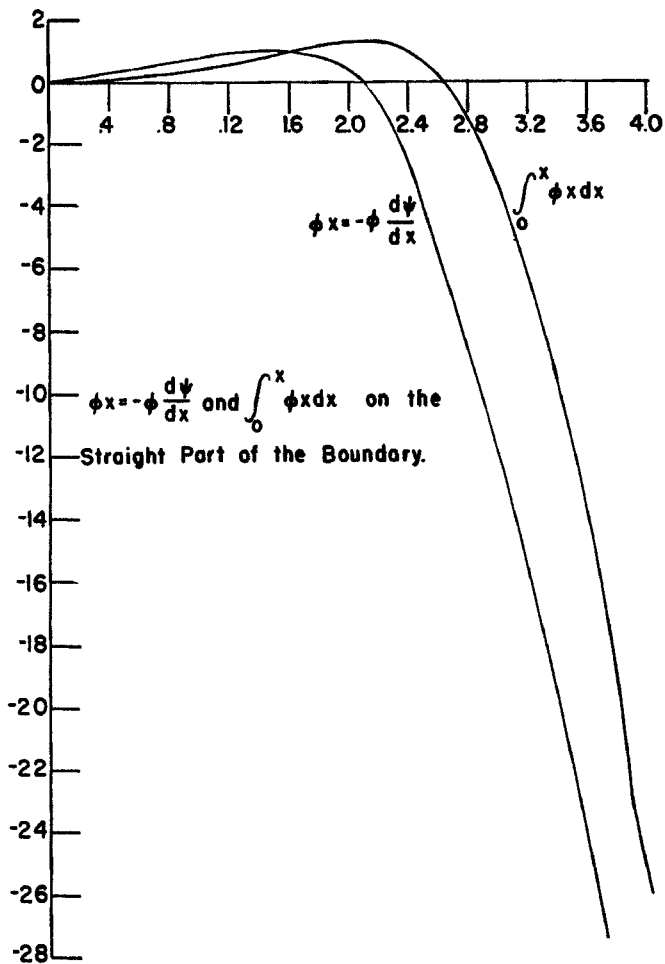
GRAPH 2b



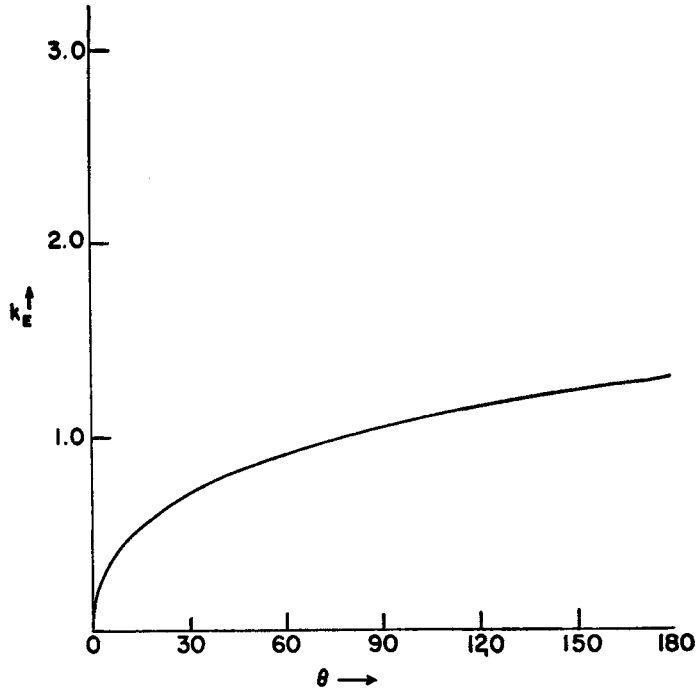
GRAPH 3



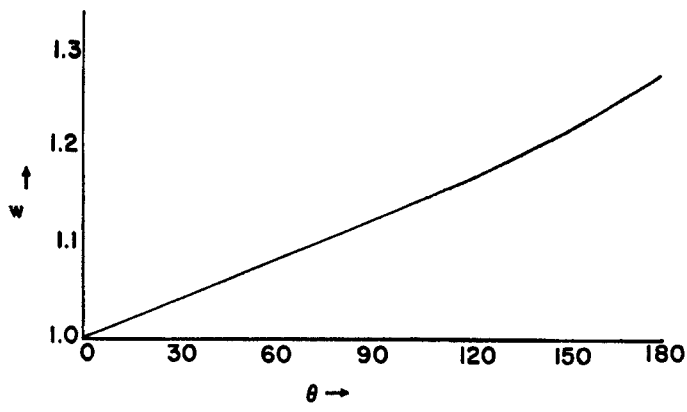
GRAPH 4



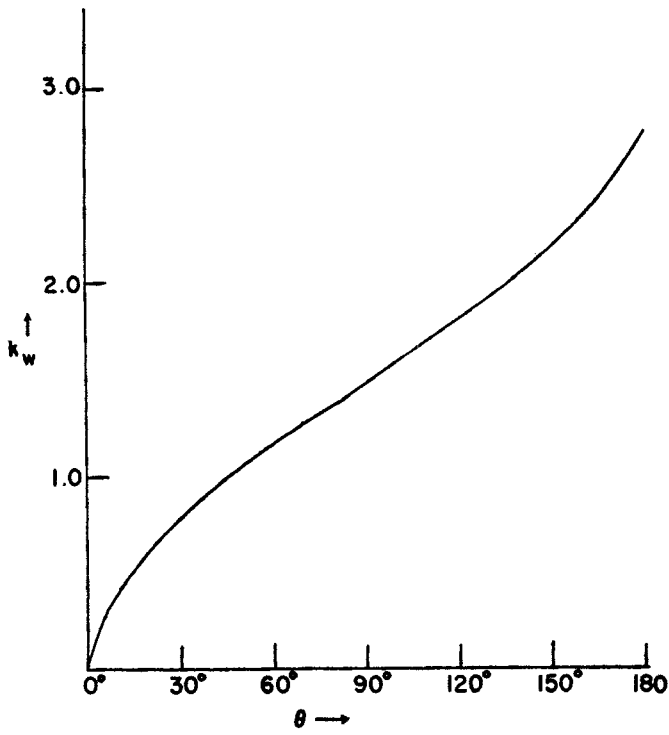
GRAPH 5



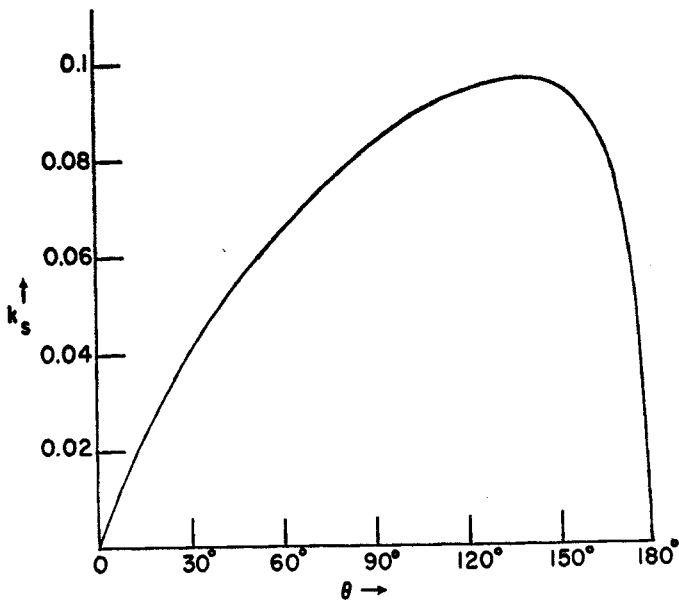
GRAPH 6



GRAPH 7



GRAPH 8



GRAPH 9

Appendix

*Measurements from Graphs on pages 160 and 161
of Watanabe's paper [5b]*

In Watanabe's notation:

$$k \cong 10^3 \frac{R}{B^2} \cdot \frac{\cot^3(\theta/2)}{6H} \cdot \frac{1}{1 - (R_x/2MH)}$$

where $R = \text{kg.}$, $B = \text{cm.}$, $M = \text{kg.}$, $H = \text{cm.}$

Graph A, $\theta = 140^\circ$, $M_0 = 1.950$ kg.

B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 114$ cm.				$H = 74$ cm.			
0.2	2.5	4.41		0.2	3	8.20 (omitted)	
0.3	4	3.14		0.3	4	4.86	
0.4	6.5	2.88		0.4	5.5	3.78	
0.5	9	2.57		0.5	7	3.09	
0.6	12	2.39		0.6	8.5	2.62	
0.7	15.5	2.30		0.7	10.5	2.39	
0.8	18.5	2.11		0.8	12	2.12	
0.9	22.0	2.01		0.9	14	1.97	
1.0	25.5	1.91		1.0	16	1.84	
1.1	29	1.83		1.1	18	1.74	
1.2	32.5	1.74		1.2	20	1.66	
1.3	36	1.68		1.3	21	1.49	
1.4	39.5	1.62		1.4	23	1.44	
			2.35	1.5	25	1.39	
				2.34			
$H = 34$ cm.							
0.4	2	2.97					
0.6	3	2.00					
0.8	4.5	1.70					
1.0	6.0	1.49					
1.2	7.5	1.32					
1.4	8.5	1.12					
1.77							

Average k for Graph A 2.24

Graph B, $\theta = 160^\circ$, $M_0 = 0.75$ kg.								
B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k	
$H = 114$ cm.				$H = 74$ cm.				
0.1	5.5	4.39	2.54	0.1	3	3.67	2.25	
0.2	14	2.81		0.2	8	2.48		
0.3	24.5	2.25		0.3	15.5	2.19		
0.4	38.5	2.09		0.4	23.5	1.96		
0.5	50.5	1.88		0.5	32.0	1.82		
0.6	64	1.81		0.6	42	1.84		
				0.7	49	1.77		
$H = 34$ cm.								
0.1	1	2.68	2.53					
0.2	4	2.71						
0.3	8.5	2.65						
0.4	15.5	2.95						
0.5	18	2.33						
0.6	22.5	2.27						
0.7	25.5	2.14						

Average k for Graph B 2.23

Graph C, $\theta = 160^\circ$, $M_0 = 1.145$ kg.							
B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 114$ cm.				$H = 74$ cm.			
0.1	5.5	4.39		0.1	3	3.61	
0.2	14	2.82		0.2	8	2.42	
0.3	25	2.25		0.3	15.5	2.13	
0.4	40	2.11		0.4	25	2.00	
0.5	53.5	1.88		0.5	32.5	1.86	
0.6	69.5	1.82		0.6	44.5	1.76	
0.7	83.5	1.74		0.7	53	1.66	
			2.43				2.21
$H = 34$ cm.							
0.1	1	2.67					
0.2	3	2.02					
0.3	6	1.82					
0.4	10	1.76					
0.5	15	1.77					
0.6	20	1.75					
0.7	23	1.58					
			1.91				

Average k for Graph C 2.18

Graph D, $\theta = 160^\circ$, $M_0 = 1.960$ kg.							
B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 114$ cm.				$H = 74$ cm.			
0.1	4	3.16	2.00	0.1	2.5	3.01	2.22
0.2	11.5	2.28		0.2	8	2.41	
0.3	22.5	2.01		0.3	17	2.31	
0.4	36	1.84		0.4	27.5	2.14	
0.5	50.5	1.70		0.5	39	2.00	
0.6	64.0	1.53		0.6	50.5	1.88	
0.7	79	1.45		0.7	61.5	1.77	
$H = 34$ cm.							
0.1	1	2.67	1.86				
0.2	3	2.01					
0.3	6	1.80					
0.4	9.5	1.64					
0.5	14.5	1.64					
0.6	20	1.63					
0.7	26	1.65					

Average k for Graph D 2.03

Graph E, $\theta = 165^\circ$, $M_0 = 0.770$ kg.							
B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 114$ cm.				$H = 74$ cm.			
0.1	15	4.99	2.82	0.1	7	3.60	2.33
0.2	35.50	3.05		0.2	19	2.50	
0.3	56.5	2.29		0.3	33	2.05	
0.4	77	1.93		0.4	47	1.80	
0.5	100	1.85		0.5	61.5	1.71	
$H = 34$ cm.							
0.1	3	3.32	2.40				
0.2	9.5	2.71					
0.3	17	2.30					
0.4	23.5	1.97					
0.5	28.5	1.72					
				Average k for Graph E			
				2.52			

Graph F, $\theta = 165^\circ$, $M_0 = 1.165$ kg.							
B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 90$ cm.				$H = 60$ cm.			
0.1	9	3.80	2.47	0.1	5	3.16	2.46
0.2	24	2.58		0.2	16	2.58	
0.3	43	2.14		0.3	30.5	2.29	
0.4	66.5	2.00		0.4	48	2.19	
0.5	86.5	1.83		0.5	64	2.09	
$H = 30$ cm.				Average k for Graph F			
0.1	2	2.55	2.09	2.34			
0.2	7	2.27					
0.3	12.5	1.86					
0.4	21	1.90					
0.5	29	1.86					

Graph G, $\theta = 165^\circ$, $M_0 = 1.975$ kg.

B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 75$ cm.				$H = 50$ cm.			
0.1	8	4.09	2.58	0.1	5	3.81	2.50
0.2	20.5	2.65		0.2	12.5	2.40	
0.3	39	2.30		0.3	26	2.29	
0.4	59	2.04		0.4	40	2.07	
0.5	78	1.83		0.5	55	1.94	
$H = 25$ cm.							
0.1	2	3.07	2.42				
0.2	6.5	2.52					
0.3	12.5	2.20					
0.4	21.5	2.25					
0.5	29	2.07					

Average k for Graph G 2.50

Graph H, $\theta = 170^\circ$, $M_0 = 0.697$ kg.

B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 100$ cm.				$H = 72$ cm.			
0.1	36	4.06	2.74	0.1	24	3.69	2.52
0.2	76	2.36		0.2	52	2.18	
0.3	111	1.79		0.3	78	1.70	
$H = 34$ cm.							
0.1	9	2.93	2.10				
0.2	21	1.84					
0.3	34	1.54					

Average k for Graph H 2.45

Graph I, $\theta = 170^\circ$, $M_0 = 1.515$ kg.

B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 100$ cm.				$H = 74$ cm.			
0.1	25.5	2.83	2.30	0.1	17	2.40	2.11
0.2	78	2.27		0.2	59	2.18	
0.3	129	1.81		0.3	98.5	1.76	
$H = 34$ cm.							
0.1	7	2.26	1.85				
0.2	20	1.66					
0.3	40.5	1.63					

Average k for Graph I 2.09Graph J, $\theta = 170^\circ$, $M_0 = 1.900$ kg.

B (cm.)	R (kg.)	k	Average k	B (cm.)	R (kg.)	k	Average k
$H = 100$ cm.				$H = 70$ cm.			
0.1	32	3.55	2.64	0.1	24	3.63	2.63
0.2	81.5	2.34		0.2	57	2.24	
0.3	147	2.04		0.3	106	2.01	
$H = 40$ cm.							
0.1	13.5	3.68	2.98				
0.2	38	2.71					
0.3	73	2.56					

Average k for Graph J 2.75

Average Values for k

140°	2.24
160°	2.21
165°	2.45
170°	2.43

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